# Three-House Demand and Solar Forecasting: Data and Evaluation

## Verified dataset facts

Ausgrid 2012-2013 household CSV: 268,557 rows, 54 fields, 300 customers, and 365 dates from 2012-07-01 through 2013-06-30. The August 2014 notes define GC as primary/general consumption excluding CL and solar; CL as controlled-load off-peak consumption; GG as gross solar generation. Interval values are kWh per half-hour, end-labeled. Generator Capacity is kWp. Blank Row Quality means actual; literal NA marks estimates/substitutes. No tariff prices are included.

## Cleaning and hourly preparation

Selected only customers with all 365 actual GC, CL and GG daily records: H001 maps to source customer 1 (3.78 kWp); H002 maps to source customer 3 (1.0 kWp); H003 maps to source customer 4 (1.0 kWp). Records were matched by customer and date. Hourly total demand is the two half-hour GC kWh values plus the matching two CL values. Solar is the corresponding two GG values, kept as a separate target. No solar generation is added to or subtracted from demand. The hourly panel contains 26,280 rows (8,760 per house), has no missing values or duplicate house-hour timestamps, and uses naive local EST/EDT wall-clock timestamps because the daily CSV provides no UTC offsets.

## Sequences, split, and model

One shared LSTM takes a 24 x 2 window of the previous demand and solar kWh values and predicts both values at the next hour. The split is chronological for each house: 80% development (7,008 hours), 20% held-out test (1,752 hours). The development period is further split chronologically into model-fit and validation portions. MinMaxScaler is fitted only on model-fit observations; validation/test do not fit the scaler. Test windows use previous observed values, including the training tail at the boundary, and are evaluated as rolling one-step forecasts. No held-out target enters training.

## Evaluation and assumptions

Training stopped after 14 epochs using early stopping. Negative outputs were clipped to zero before metrics (demand: 130; solar: 1136). The persistence baseline copies the latest observed hour. Demand LSTM overall MAE is worse than persistence in this run; solar metrics improve, so this is a baseline needing further evaluation, not a validated deployment model. Price scenarios are illustrative hypothetical values only and must be replaced with a verified tariff before economic analysis.

## Held-out metrics in kWh

| Scope | Model | Target | N | MAE | RMSE |
|---|---|---|---|---|---|
| ALL | shared_lstm_24h | total_demand | 5256 | 0.4953 | 0.8026 |
| ALL | shared_lstm_24h | solar_generation | 5256 | 0.0730 | 0.1439 |
| H001 | shared_lstm_24h | total_demand | 1752 | 0.5718 | 0.8641 |
| H001 | shared_lstm_24h | solar_generation | 1752 | 0.1279 | 0.2240 |

| Scope | Model | Target | N | MAE | RMSE |
|---|---|---|---|---|---|
| H002 | shared_lstm_24h | total_demand | 1752 | 0.4097 | 0.5716 |
| H002 | shared_lstm_24h | solar_generation | 1752 | 0.0446 | 0.0773 |
| H003 | shared_lstm_24h | total_demand | 1752 | 0.5045 | 0.9268 |
| H003 | shared_lstm_24h | solar_generation | 1752 | 0.0466 | 0.0775 |
| ALL | persistence_last_hour | total_demand | 5256 | 0.4310 | 0.8557 |
| ALL | persistence_last_hour | solar_generation | 5256 | 0.0864 | 0.2001 |
| H001 | persistence_last_hour | total_demand | 1752 | 0.5496 | 0.8175 |
| H001 | persistence_last_hour | solar_generation | 1752 | 0.1609 | 0.3192 |
| H002 | persistence_last_hour | total_demand | 1752 | 0.2088 | 0.5595 |
| H002 | persistence_last_hour | solar_generation | 1752 | 0.0517 | 0.0978 |
| H003 | persistence_last_hour | total_demand | 1752 | 0.5347 | 1.1023 |
| H003 | persistence_last_hour | solar_generation | 1752 | 0.0467 | 0.0929 |